

Komponen	Spesifikasi
DevKit ESP32-WROOM-32	30-pin, USB-C, onboard LDO 3.3V
Panel Surya	6V 10W, monocrystalline
Baterai Li-ion	18650 3.7V 3400mAh, dengan BMS dan holder
BFC Buck-Boost	Adjustable XL4016, input 3-30V, output 1-28V 3A
Modul E-Step HV	10-30kV DC, 12V input, arus <50mA
Kawat HV	Silicone 30kV, 1-2mm ² , merah
Jarum Emitter Corona	Stainless 0.3-0.5mm, runcing, 20-40 jarum
Plat/Kisi Pengumpul	Stainless 316L perforated, 2-4mm, 300x400mm
BME280	I2C, sensor suhu + kelembapan + tekanan
Anemometer	Cup 0-25 m/s, output pulse
Sensor Kelembapan Kapasitif	Grove capacitive humidity, untuk deteksi kabut
Relay Module Opto-Isolated	2-channel, 5V coil, 10A kontak, SRD-05VDC-SL-C
Layar OLED	0.96 inch SSD1306 128x64 I2C
Push Button E-STOP	22mm latching, kontak NC
Ground Rod	Tembaga 8-10mm, 1m
Dioda Flyback	1N4007, 1A 1000V
Fuse	10A, automotive blade, dengan holder
Main Switch	SPST 10A, panel mount
Kipas Pendingin	40mm 12V, plus heat sink aluminium
Pre-Filter	100-mesh stainless
HV Probe	1k Ω $\times$ 30kV, untuk pengukuran tegangan HV
Ammeter DC	0-100mA, untuk memastikan arus HV aman